

Implementation and Evaluation of a Model Transfer and Integration Workflow for Distributed Spiking Neural Network Learning

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1. Introduction

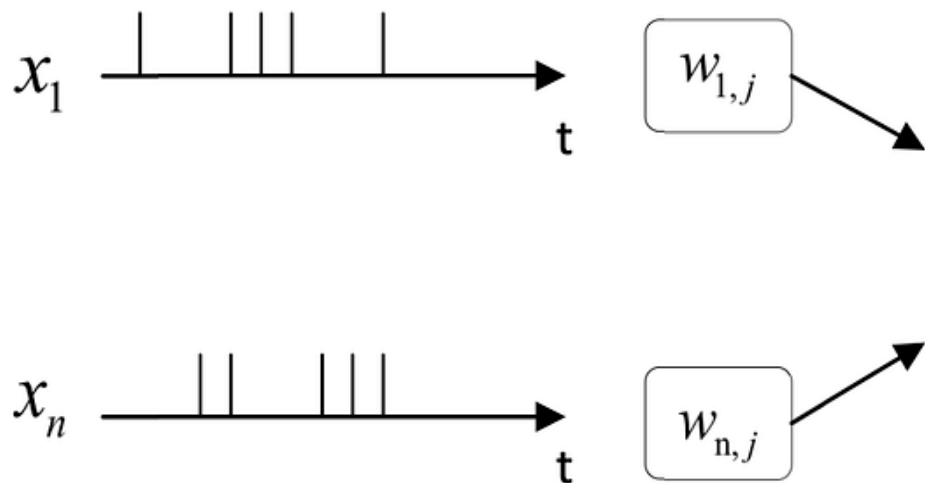
2. Proposal

3. Evaluation

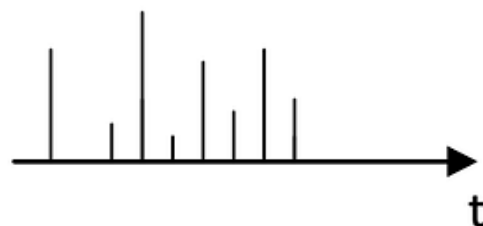
4. Conclusion

Spiking Neural Networks

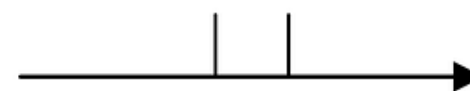
INPUT SPIKES



WEIGHTED INPUT SPIKES



OUTPUT SPIKES



x_i : input spike
 $w_{i,j}$: connection weight
 t : time
 j : spiking neuron

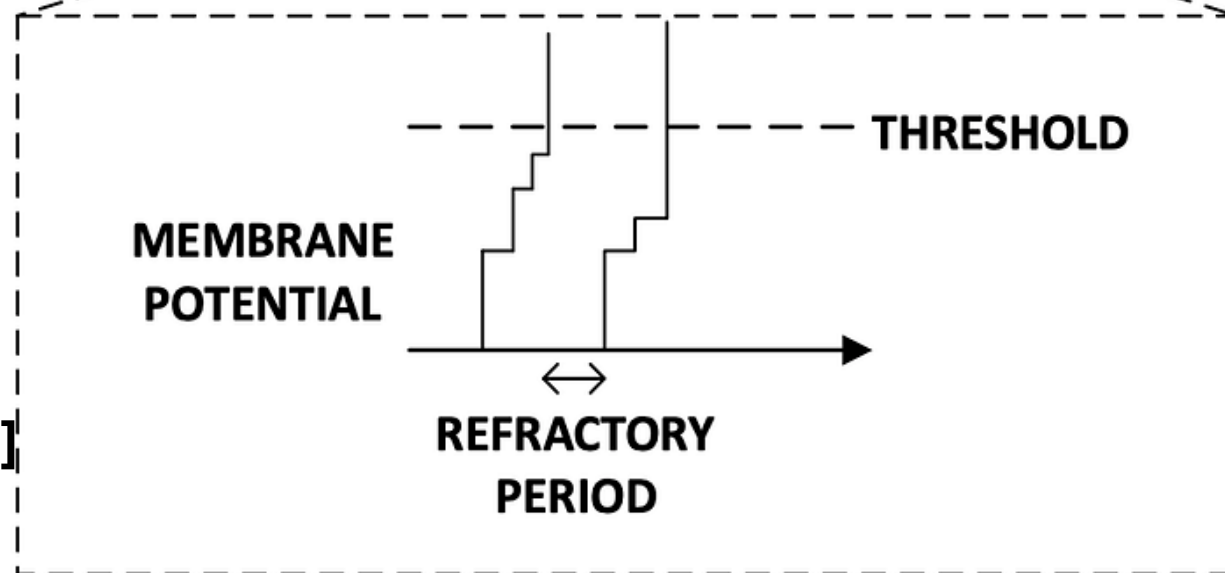
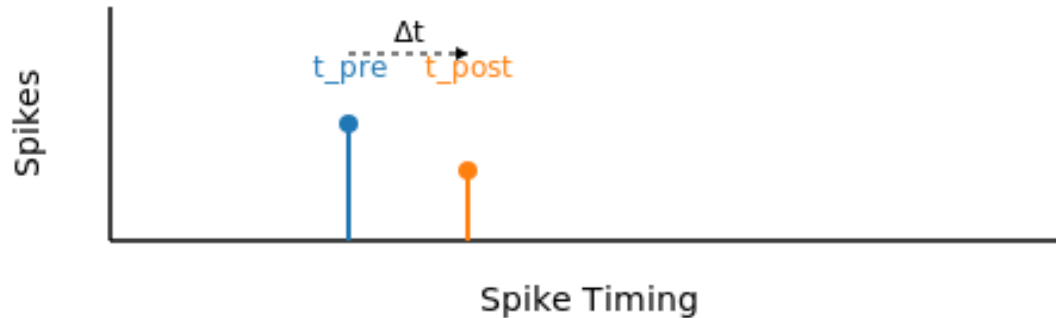


Image $\rightarrow$ spike trains $\rightarrow$ output spikes [1]

Spike-Timing-Dependent Plasticity

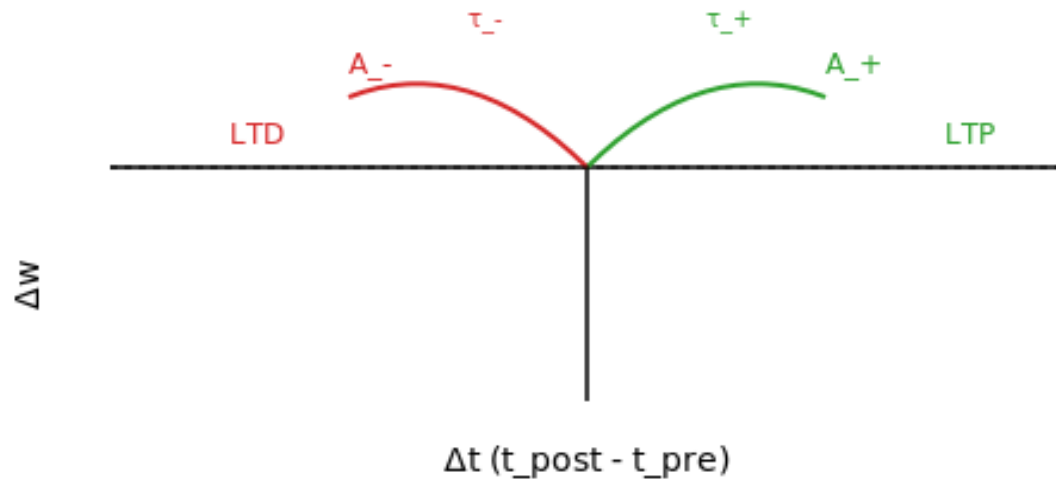


Basic relation

$$\Delta t = t_{post} - t_{pre}$$

$$\Delta w = A_+ \exp(-\Delta t / \tau_+) \quad \text{if } \Delta t > 0$$

$$\Delta w = -A_- \exp(\Delta t / \tau_-) \quad \text{if } \Delta t < 0$$



**Pre → Post
weight increases**

**Post → Pre
weight decreases**

Learning depends on spike timing [2]

Symbols

Δt : spike timing difference

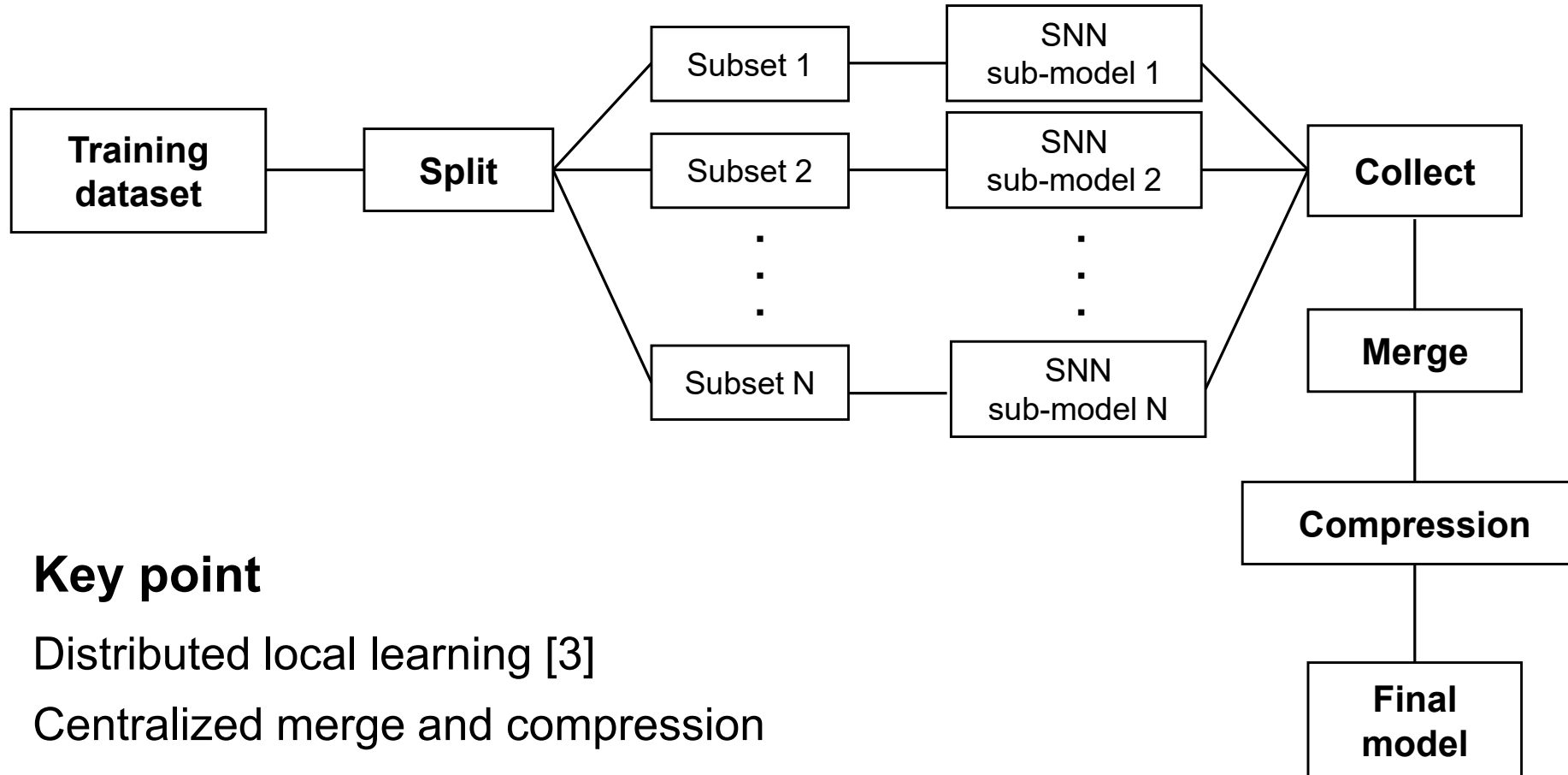
t_{pre} , t_{post} : pre-/post-synaptic spike times

Δw : change in synaptic weight

A_+ , A_- : max increase/decrease

τ_+ , τ_- : time constants

EnsembleSTDP



Key point

Distributed local learning [3]

Centralized merge and compression

Research Motivation

- Parallel sub-model training can reduce training time. [3]
- But model-related file transfer, delayed senders, and receiver-side processing may reduce this benefit.
- This study evaluates workflow completion time and final merged-model accuracy.

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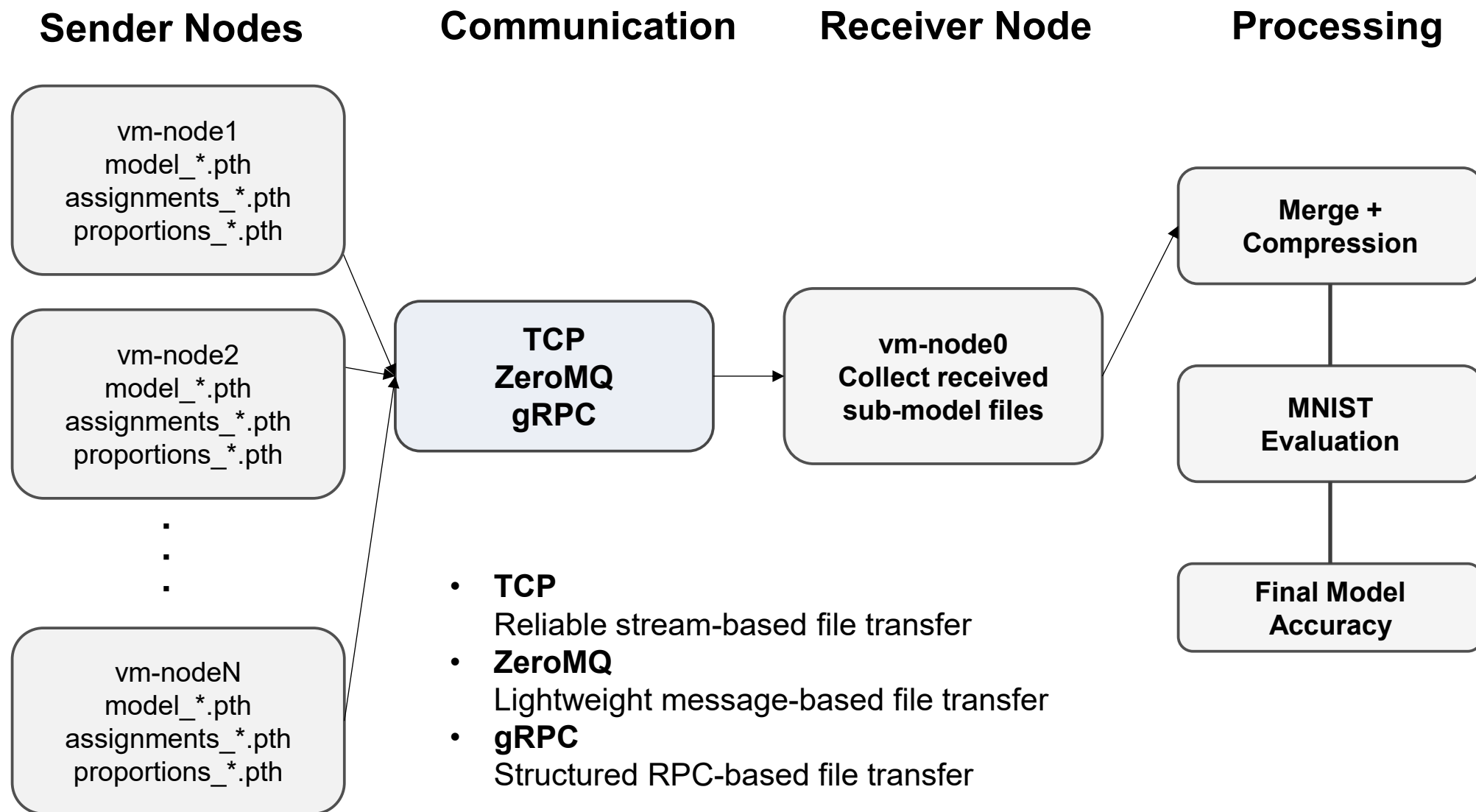
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Overall Experimental Setup



Sub-model and Quality Evaluation

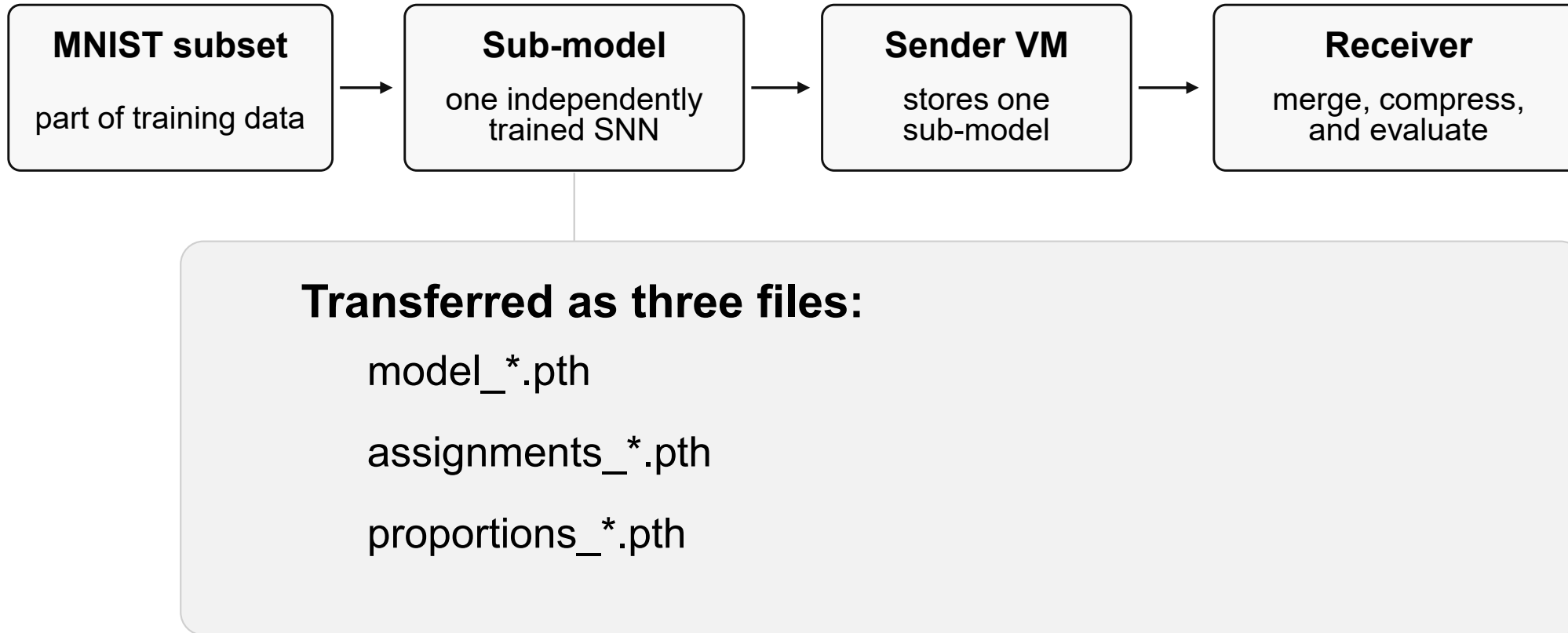


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Evaluation Objectives

What to Evaluate

Why

- **Workflow feasibility**
→ Model Transfer and Integration
- **Workflow completion time**
→ Sequential vs. Concurrent
→ Controlled Straggler
→ Arrival Skew
- **Receiver-side processing**
→ Bottleneck Breakdown
- **Final merged-model accuracy**
→ Same accuracy across protocols
→ Impact of a Low-Accuracy Sub-model

Confirm successful transfer and integration

Evaluate execution mode and sender timing

Identify the dominant bottleneck

Examine protocol and sub-model quality

Experimental Environment

Environment: Ubuntu VMs / VirtualBox

Receiver: vm-node1

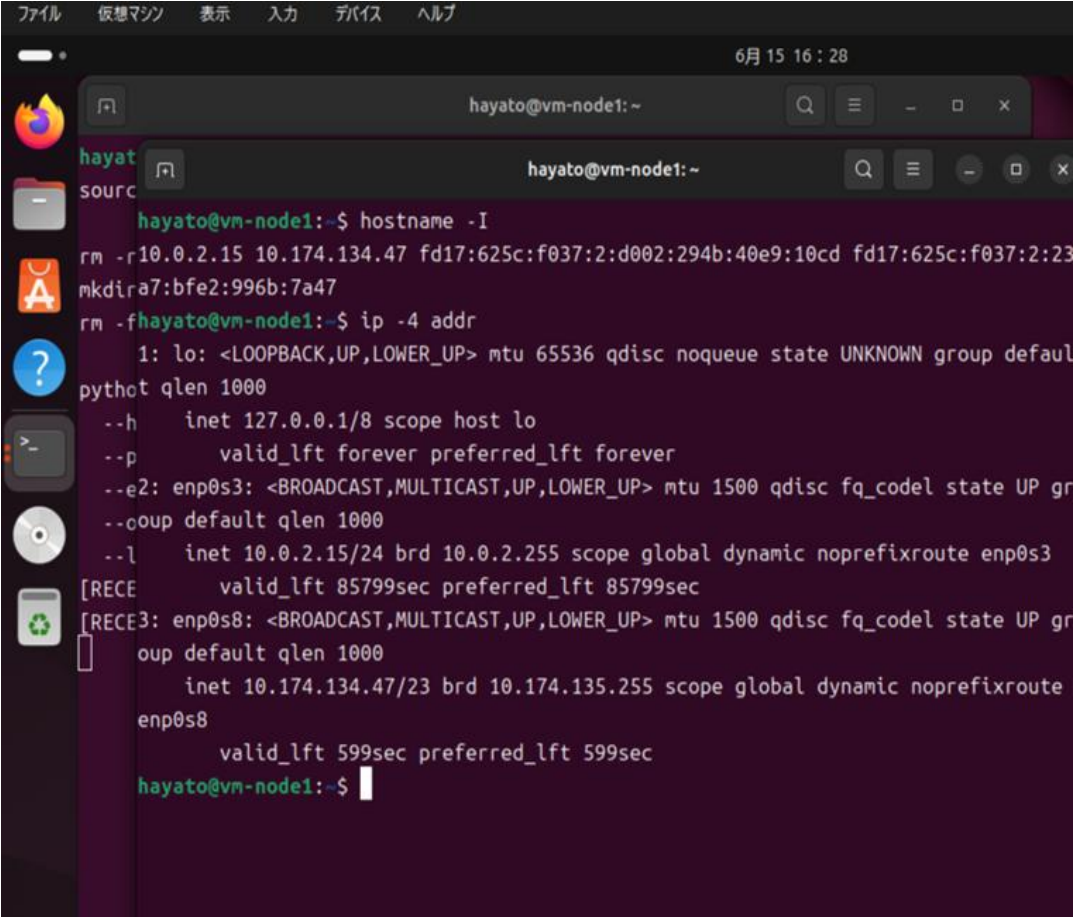
Senders: up to 9 sender VMs

Protocols: TCP / ZeroMQ / gRPC

Dataset: MNIST test set (10,000)

Framework: BindsNET [4]

Sender VMs → Receiver VM → Merge /
Compression → MNIST Evaluation

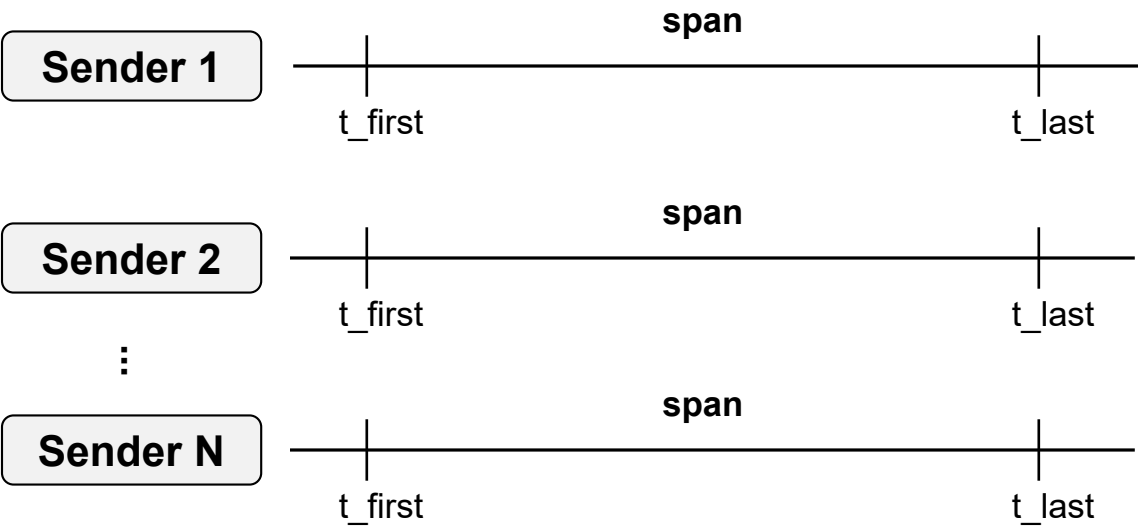


```
hayato@vm-node1:~$ hostname -I
10.0.2.15 10.174.134.47 fd17:625c:f037:2:d002:294b:40e9:10cd fd17:625c:f037:2:23
mktira7:bfe2:996b:7a47
hayato@vm-node1:~$ ip -4 addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
    link-loopback (00:00:00:00:00:00)
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group
    default qlen 1000
    link ether 08:00:27:00:00:00
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 85799sec preferred_lft 85799sec
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group
    default qlen 1000
    link ether 08:00:27:00:00:00
    inet 10.174.134.47/23 brd 10.174.135.255 scope global dynamic noprefixroute enp0s8
        valid_lft 599sec preferred_lft 599sec
hayato@vm-node1:~$
```

vm-node1 running in Oracle VirtualBox

Proposed Timing Measurement Method

Receiver-side timestamps



Corrected transfer time
= Σ sender span

$T_{corrected} =$
sum over senders (t_last,sender - t_first,sender)

Removes manual idle time
Not pure network latency

Concurrent execution metrics

Global span
Arrival skew

Global span: first received file → last received file
Arrival skew: earliest sender arrival → latest sender arrival

Result 1: Workflow Feasibility

Purpose: Verify workflow feasibility and final model correctness.

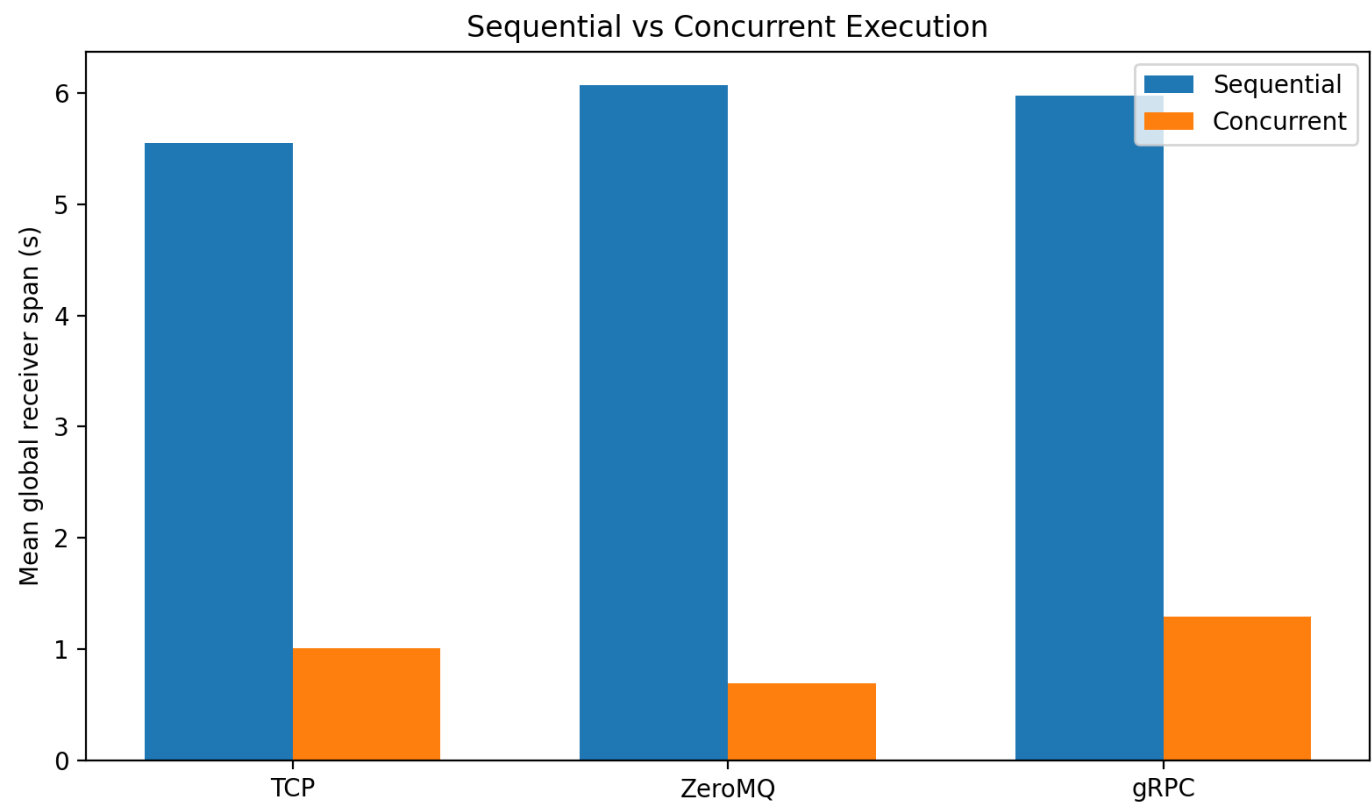
Configuration	Final neurons	Accuracy
3-sender asymmetric	300	0.8676
9-sender symmetric	800	0.8800
9-sender asymmetric, initial	1000	0.5719
9-sender asymmetric, improved	1000	0.8232

3-sender asymmetric: TCP = ZeroMQ = gRPC = 0.8676

- TCP, ZeroMQ, and gRPC successfully transferred the required model-related files.

Result 2-A: Sequential vs Concurrent Execution

Purpose: Determine whether concurrent execution reduces workflow completion time.

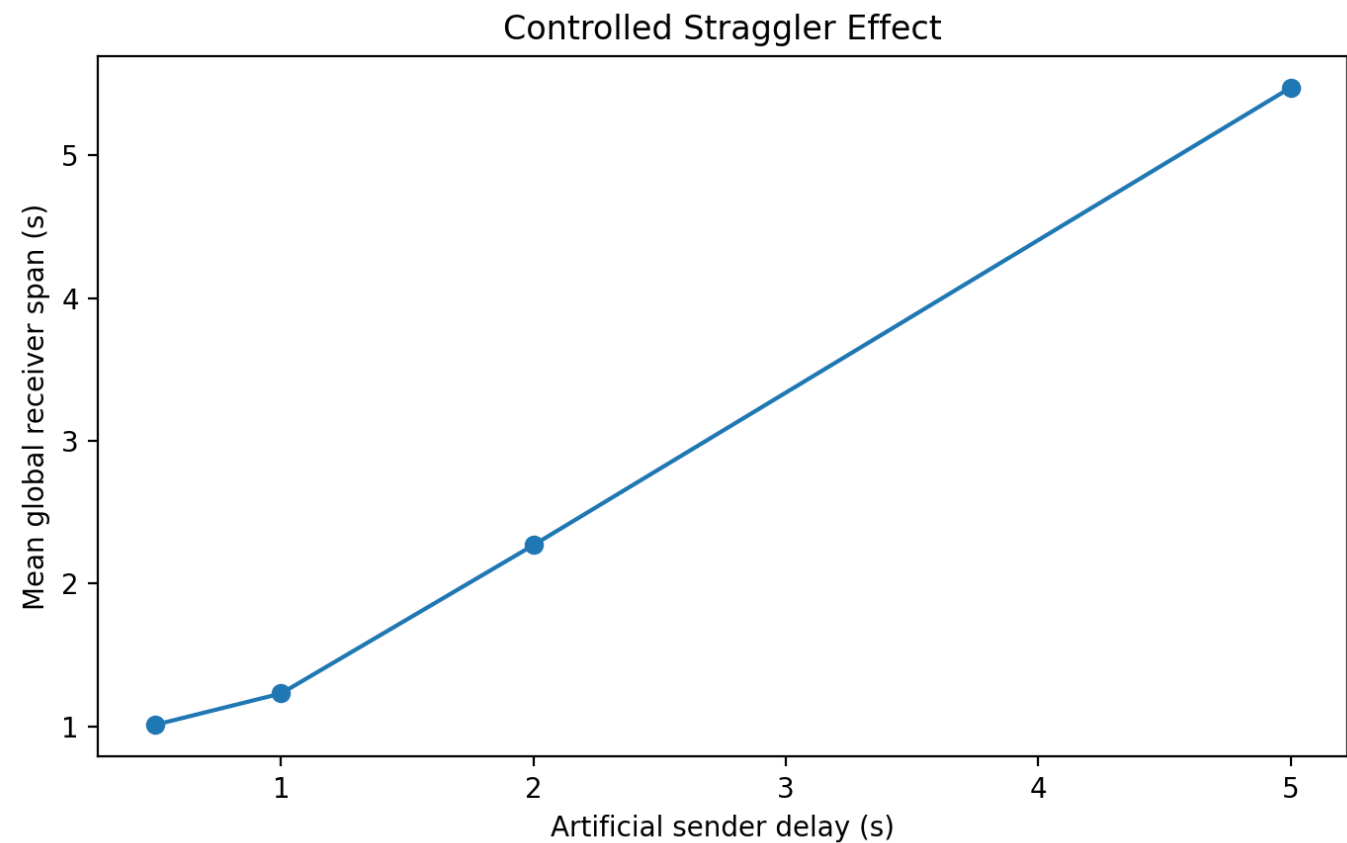


Approach	Reduction
TCP	81.90%
ZeroMQ	88.64%
gRPC	78.37%

- The reduction ranged from **78.37%** to **88.64%**.
- Sender execution mode has a large impact on the overall completion time.

Result 2-B: Controlled Straggler Effect

Purpose: Examine whether one delayed sender determines global completion time.

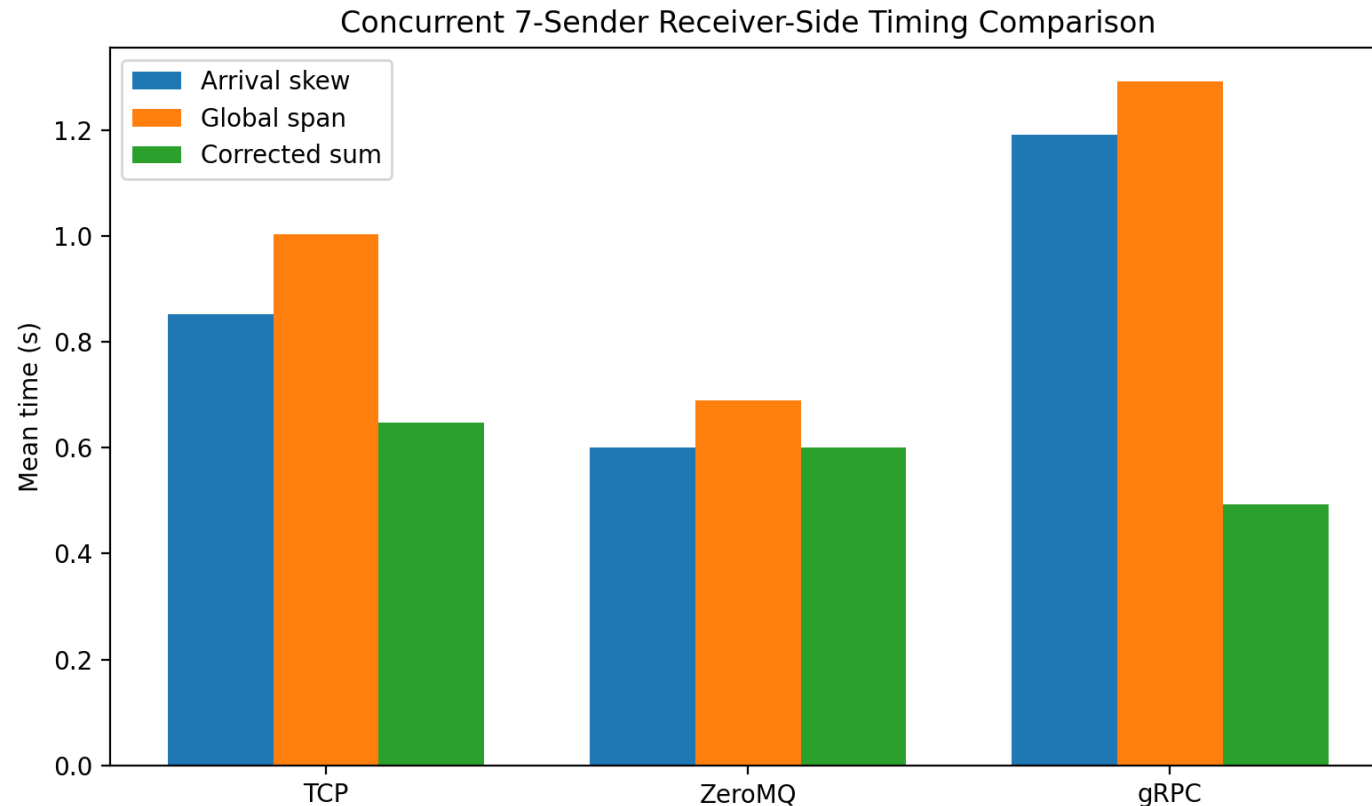


Delay	Global span
0.5 s	1.009 s
1.0 s	1.229 s
2.0 s	2.269 s
5.0 s	5.475 s

- Delay added to one sender
- Delay $\uparrow \rightarrow$ global span $\uparrow$
- One straggler can determine completion time

Result 2-C: Arrival Skew in Concurrent Execution

Purpose: Evaluate the effect of sender arrival imbalance on global completion.



gRPC:

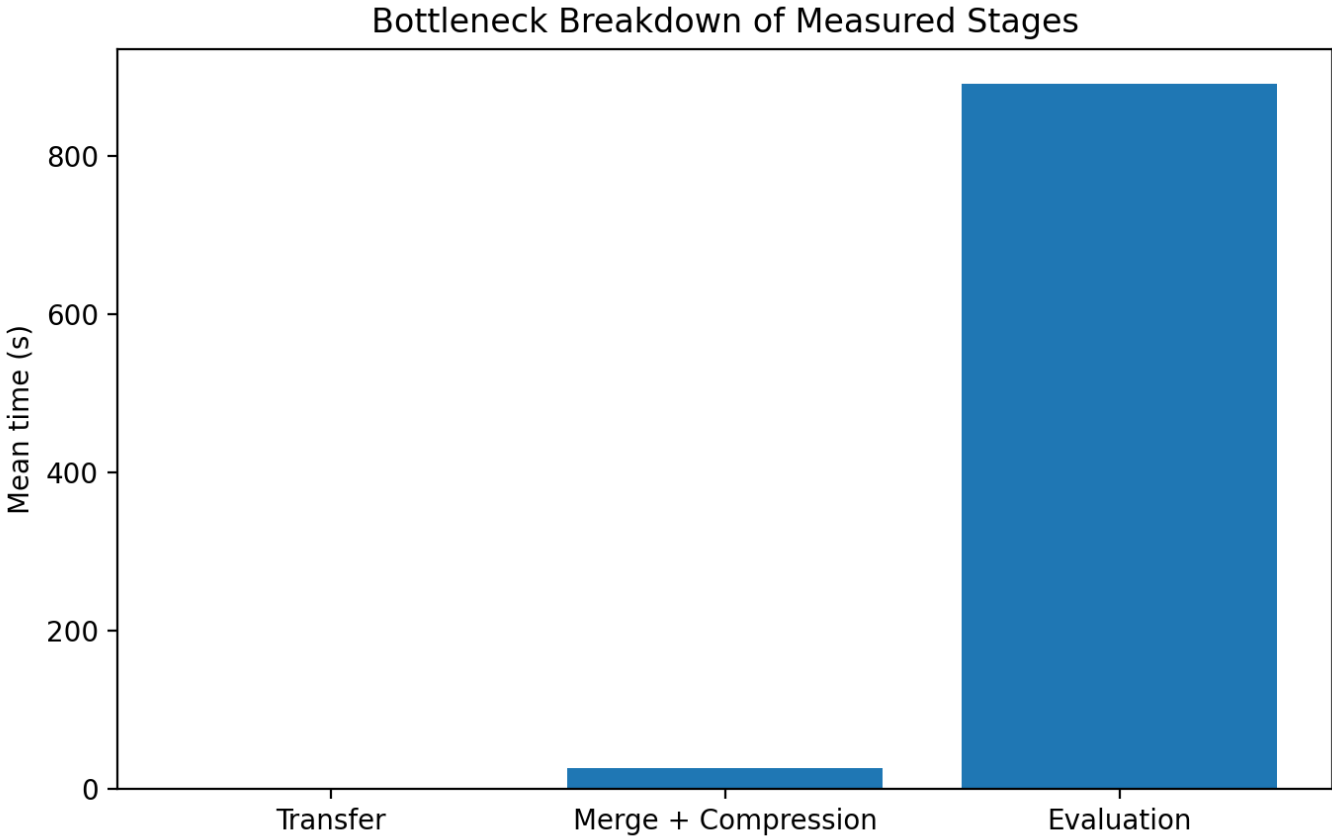
- per-sender span: 0.070 s
- arrival skew: 1.193 s
- global span: 1.293 s

Finding

- Short sender span $\neq$ short global span
- Arrival skew increases completion time

Result 3: Receiver-Side Bottleneck

Purpose: Identify the dominant receiver-side processing stage.



Stage	Mean time
Transfer	1.26 s
Merge + compression	26.23 s
Evaluation	890.75 s

Evaluation accounted for about 97% of the measured workflow time.

Result 4: Final Model Quality

Impact of a Low-Accuracy Sub-model

Check	Condition	Accuracy
Original model_1	250 neurons, 0-6000	0.5349
Remove model_1	model_2-9 only	0.8033
Change data range	250 neurons, 6000-12000	0.7358
Change neurons	200 neurons, 0-6000	0.8290

- These results indicate that **sub-model quality**, rather than the communication approach, was a major factor in the low asymmetric-model accuracy.

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Conclusion and Future Work

Conclusion

- Implemented and validated the model-transfer and receiver-side integration workflow using TCP, ZeroMQ, and gRPC.
- Workflow performance depended on sender coordination, receiver-side processing, and sub-model quality—not on communication approach alone.

Future work

- Evaluate the workflow on physical and heterogeneous devices.
- Improve sender coordination and reduce receiver-side processing cost.

References

- [1] W. Maass, “Networks of spiking neurons: The third generation of neural network models,” *Neural Networks*, vol. 10, no. 9, pp. 1659–1671, 1997.
- [2] H. Markram, W. Gerstner, and P. J. Sjöström, “Spike-Timing-Dependent Plasticity: A Comprehensive Overview,” *Frontiers in Synaptic Neuroscience*, vol. 4, Art. 2, 2012.
- [3] Y. Hanyu and K. N. Dang, “EnsembleSTDP: Distributed in-situ Spike Timing Dependent Plasticity Learning in Spiking Neural Networks,” 2024 IEEE 17th International Symposium on Embedded Multicore/Many-core Systems-on-Chip (MCSoc), pp. 434–440, 2024.
- [4] H. Hazan et al., “BindsNET: A Machine Learning-Oriented Spiking Neural Networks Library in Python,” *Frontiers in Neuroinformatics*, vol. 12, Art. 89, 2018.

Research group GitHub: <https://github.com/klab-aizu>

EnsembleSTDP source code: <https://github.com/klab-aizu/EnsembleSTDP>

Thank you for your attention!

Any questions?